WORKED SOLUTION

STEP I 1991, Question 13

Mechanics • Conservation of Momentum

STEP I 1991, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q13

The key idea

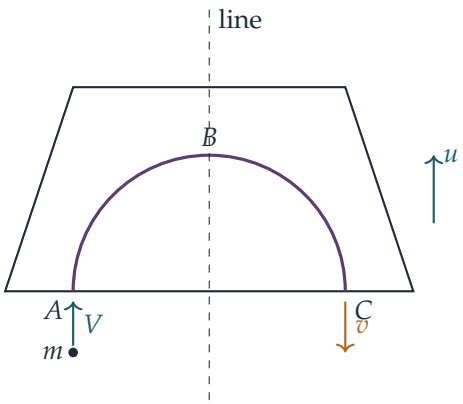
The table is smooth and the groove is smooth, so nothing outside the system pushes along the line, and no force does any work. That gives us two conservation laws: momentum along the line and kinetic energy. The groove's tangent at the exit is parallel to the line, so at the exit *both* velocities lie along the line, and the two laws then work just like a one-dimensional elastic collision. For the fixed lamina, the average force is simply the total impulse divided by the time taken.

A note on the wording: the question says the tangents at A and B are parallel to the line, but the diagram shows that it is the tangents at the two ends, A and C, that are parallel to it (at B the groove is at right angles to the line). We use A and C, which is clearly what is intended.

Solution

The moving lamina

Model the particle as a point mass, the lamina as a rigid body that can only translate along the line, and all contacts as smooth. Take the line as an axis, with the positive direction being the particle's direction of motion when it enters at A.



Let the particle leave C with velocity v along the line (positive in the direction of V), and let the lamina then have velocity u along the line.

Ask yourself Why must the particle's velocity at C be along the line?

Relative to the lamina, the particle moves along the groove, so its relative velocity at C is along the tangent there, which is parallel to the line. The lamina itself only moves along the line. Adding the two, the particle's actual velocity at C is along the line as well. So at the exit we only need one number for each body.

During the motion, the forces on the system (particle plus lamina) from outside are the table's normal reaction, which is vertical, and whatever keeps the lamina on its line, which is perpendicular to the line (the line is smooth). The groove's force on the particle and the particle's force on the groove are internal and cancel in pairs. So momentum along the line is conserved:

$$mV = mv + Mu. \tag{1}$$

Every contact is smooth, and the table is horizontal, so no work is done and kinetic energy is conserved. (The normal reaction of the groove does work on the particle and on the lamina separately, but these two amounts cancel, because the reaction is perpendicular to the *relative* velocity.) So

$$\tfrac{1}{2}mV^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}Mu^2. \tag{2}$$

Rewrite (1) and (2) as

$$m(V - v) = Mu, \qquad m(V - v)(V + v) = Mu^2.$$

One solution is $v = V$, $u = 0$: this is the situation at A, before anything has happened. At C the particle is moving *backwards* relative to the lamina, so $v \neq V$ and $u \neq 0$, and we may divide the second equation by the first to get

$$V + v = u.$$

(This says the relative speed of the particle with respect to the lamina is V at both ends, just as in an elastic collision.) Substituting $u = V + v$ into (1):

$$mV = mv + MV + Mv \quad \Longrightarrow \quad v = -\frac{M - m}{M + m}V, \qquad u = V + v = \frac{2m}{M + m}V.$$

Since $m < M$, v is negative: the particle comes out of C moving back the way it came. So when the particle leaves the groove,

$$\text{speed of particle} = \frac{M-m}{M+m}\,V, \qquad \text{speed of lamina} = \frac{2m}{M+m}\,V,$$

with the lamina moving in the original direction of the particle and the particle moving the opposite way.

The particle does get round the groove: at any point where it has stopped relative to the lamina, it would share the lamina's velocity, and then the kinetic energy would be $\frac{1}{2}(mV)^2/(M+m) < \frac{1}{2}mV^2$, which energy conservation forbids. So its relative speed never drops to zero.

The lamina held fixed

Now the lamina cannot move. The groove's reaction on the particle is perpendicular to its velocity, so it does no work, and the particle goes round the semicircle at constant speed V. The semicircle has length πr, so the time in the groove is

$$T = \frac{\pi r}{V}.$$

The particle's momentum changes from mV forwards to mV backwards, so the total impulse the lamina gives the particle is $2mV$, backwards along the line. Its momentum perpendicular to the line is zero at both ends, so the perpendicular part of the impulse is zero.

By Newton's third law the particle gives the lamina an impulse $2mV$ forwards along the line. The lamina stays at rest, so the peg must supply the opposite impulse, and the lamina pushes on the peg with total impulse $2mV$ in the direction of the particle's original motion. (Any sideways push from the line averages to zero, by the previous paragraph.) Dividing by T, the average force on the peg is

$$\frac{2mV}{\pi r/V} = \frac{2mV^2}{\pi r},$$

in the direction in which the particle entered the groove. (The question's "average force per unit time" means exactly this: impulse per unit time.)

We can also see this directly. The particle needs a force mV^2/r towards the centre, so it pushes on the lamina with mV^2/r outwards. When the radius to the particle makes angle θ with CA (where θ runs from 0 to π at a steady rate), the component of this push along the line is $(mV^2/r)\sin\theta$, and the average of $\sin\theta$ over $[0,\pi]$ is $2/\pi$.

Checks

- If $M \to \infty$ the lamina barely moves: $u \to 0$ and the particle's speed tends to V, as for a fixed groove.
- If $m = M$ the particle comes out at rest and the lamina takes all its speed V, like an elastic collision between equal masses.

- With $M = 3$, $m = 1$, $V = 2$: $v = -1$ and $u = 1$. Momentum: $1 \cdot (-1) + 3 \cdot 1 = 2 = mV$. Energy: $\frac{1}{2} + \frac{3}{2} = 2 = \frac{1}{2} \cdot 1 \cdot 2^2$. A numerical simulation of the equations of motion (for a semicircular groove and for a half-elliptical one) gave the same values to six decimal places, and numerical integration of $(mV^2/r)\sin\theta$ for the fixed lamina gave average force $2mV^2/(\pi r)$, e.g. 1.5117 for $m = 0.7$, $r = 1.3$, $V = 2.1$.

Common slips

- Keeping the root $v = V$, $u = 0$: it satisfies both equations but describes the moment of entry, not exit.
- Using the maximum force mV^2/r instead of the average along the line; the sideways components cancel, and the along-line component reaches mV^2/r only at B.

Beyond STEP

The first part is a hidden collision: whatever the shape of the groove, only the conserved quantities and the exit direction matter, which is the idea behind treating scattering problems by conservation laws without knowing the forces. In Lagrangian mechanics, the coordinate along the line does not appear in the Lagrangian, so its conjugate momentum is conserved automatically; this is the start of the link between symmetries and conservation laws (Noether's theorem). Kleppner and Kolenkow's *An Introduction to Mechanics* treats momentum, impulse and collisions in this spirit.

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